Existence and regularity of the minima of integral shape functionals

Francesco Paolo Maiale

Gran Sasso Science Institute (GSSI), L'Aquila, Italy
francescopaolo.maiale@gssi.it

Joint work with G. Buttazzo (University of Pisa), D. Mazzoleni (University of Pavia), G. Tortone (University of Pisa), and B. Velichkov (University of Pisa)



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Outline

1. Preliminaries
2. Existence of optimal shapes
3. Regularity of optimal shapes
4. Free boundary systems

A **shape optimization problem** is a variational problem of the form

$$\min\{\mathcal{F}(\Omega) \,:\, \Omega \in \mathcal{A}\}.$$

The functional $\mathcal{F}(\Omega)$ depends on the solution of a PDE in Ω, e.g.

$$\mathcal{F}(\Omega) = \int_\Omega |\nabla u|^2 \, dx$$

where u is a solution to the eigenvalue problem

$$-\Delta u = \lambda u \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega, \qquad \int_\Omega u^2 \, dx = 1.$$

Additional terms like the **perimeter** or **measure** of Ω are often involved.

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$$\min\{\mathcal{F}(\Omega) \ : \ \Omega \in \mathcal{A}\}.$$

The family of admissible sets $\mathcal{A}$ can be:

- **all open** (or quasi-open) **sets** contained in $D \subseteq \mathbb{R}^d$ or in $\mathbb{R}^d$ itself;
- **rectifiable sets with** $\mathcal{H}^{d-1}(\partial\Omega) \leq L$ contained in $D \subseteq \mathbb{R}^d$ or $\mathbb{R}^d$;
- **all sets of measure** $\leq m$ contained in $D \subseteq \mathbb{R}^d$ or $\mathbb{R}^d$;
- **all convex sets** contained in $D \subseteq \mathbb{R}^d$;
- **all sets with Lipschitz boundary** (or C^1) contained in $D \subseteq \mathbb{R}^d$.

Spectral functionals. Minimize, among all $\Omega \subseteq D$ with $|\Omega| \leq 1$,

$$\mathcal{F}(\Omega) = F(\lambda_1(\Omega), \dots, \lambda_k(\Omega)),$$

where $\lambda_i(\Omega)$ is the i-th Dirichlet eigenfunction.

- **Existence of optimal shapes**
  - robust existence theory (Buttazzo and Dal Maso 1993b), if the box D is a bounded set;
  - existence in $\mathbb{R}^d$ (Bucur 2012; Mazzoleni and Pratelli 2013).
- **Regularity of the free boundaries**
  - $\lambda_1(\Omega)$ and D bounded (Russ, Trey, and Velichkov 2019);
  - $\lambda_1(\Omega) + \cdots + \lambda_k(\Omega)$ (Mazzoleni, Terracini, and Velichkov 2017);
  - $\lambda_k(\Omega)$ and more general functionals (Kriventsov and F. Lin 2019).

Integral functionals. Minimize, among all $\Omega \subseteq D$ with $|\Omega| \leq 1$,

$$\mathcal{F}(\Omega) = \int_\Omega j(x, u_\Omega)\, dx,$$

where u_Ω is the solution of some PDE on Ω. This is typical in *optimal control* or *models in physics and engineering*. For example, when

$$-\Delta u_\Omega = f(x) \quad \text{in } \Omega, \qquad u_\Omega = 0 \quad \text{on } \partial\Omega,$$

there are several **existence results**; for instance, when:

- j monotone and D bounded (Buttazzo and Dal Maso 1993a)
- j ϵ-perturbation of monotone function (Buttazzo and Velichkov 2016)
- in general, there are counterexamples (Bucur and Buttazzo 2005)

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Let $D \subset \mathbb{R}^d$ be an open set of finite measure. For $\Omega \subseteq D$ and $p > 1$, we consider the following PDE:

$$-\Delta_p u_\Omega = f \quad \text{in } \Omega, \qquad u_\Omega = 0 \quad \text{in } \overline{D} \setminus \Omega. \tag{1}$$

In Buttazzo, Maiale, and Velichkov (2021), we focus on optimization problems of the form

$$\min \{\mathcal{F}(\Omega) \ : \ \Omega \text{ open}, \ \Omega \subseteq D\}. \tag{2}$$

For simplicity, here we consider the **affine functional**:

$$\mathcal{F}(\Omega) = -\int_D g(x) u_\Omega \, dx + |\Omega|. \tag{3}$$

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Theorem (Buttazzo, M., Velichkov)

Let $f \geq 0$ be a function in $L^q(D)$, $q > d/p$ and $q \geq 1$, and let $g \geq 0$ be measurable function in $L^r(D)$ for some $r > 1$. Then:

- *if there is $C > 0$ such that $f(x) \leq Cg(x)$ for every $x \in D$, then there is an open set $\bar{\Omega} = \{\bar{u} > 0\} \subset D$ solution to (2);*
- *any optimal shape $\bar{\Omega}$ has finite perimeter.*

(a) When f and g are **proportional**, then $\bar{\Omega} = \{\bar{u} > 0\}$ and $\bar{u}$ can be characterized as the solution to a variational problem:

$$\min_{u \in W_0^{1,p}(D)} \left\{ \frac{1}{p} \int_D |\nabla u|^p \, dx - \int_D f(x) u \, dx + \lambda \frac{p-1}{p} \, |\{u > 0\}| \right\}.$$

The regularity of the free boundary $\partial\{\bar{u} > 0\}$ is obtained, for instance, in Danielli and Petrosyan (2005) when $f = 0$.

(b) When f and g are **not proportional**, given Ω optimal shape, the corresponding state function u_Ω is a priori optimal only among functions that serve as state functions on other domains.

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Problem. Minimize the functional $\mathcal{F}(\Omega) = \int_\Omega -g(x)u_\Omega\,dx + |\Omega|$ among all sets $\Omega \subseteq D$ open, where:

$$-\Delta u_\Omega = f(x) \quad \text{in } \Omega, \qquad u_\Omega = 0 \quad \text{on } \partial\Omega.$$

Theorem (Buttazzo, Maiale, Mazzoleni, Tortone, and Velichkov (2024) and Maiale, Tortone, and Velichkov (2023))

Assume that $0 \le cf(x) \le g(x) \le Cf(x)$ *for some* $0 < c \le C$, *and suppose that* $f, g \in L^1 \cap L^\infty \cap C^2(D)$. *Then any optimal shape* $\Omega \subseteq D$ *satisfies*

$$\partial\Omega \cap D = \mathrm{Reg}(\partial\Omega) \cup \mathrm{Sing}(\partial\Omega),$$

where:

1. $\mathrm{Reg}(\partial\Omega)$ *is locally* $C^{1,\alpha}$ *for some* $\alpha \in (0,1)$;
2. $\mathrm{Sing}(\partial\Omega)$ *has Hausdorff dimension* $\le d - d^*$, *with* $d^* \in \{5,6,7\}$.

Problem. Minimize the functional $\mathcal{F}(\Omega) = \int_\Omega -g(x)u_\Omega \, dx + |\Omega|$ among all sets $\Omega \subseteq D$ open, where:

$$-\Delta u_\Omega = f(x) \quad \text{in } \Omega, \qquad u_\Omega = 0 \quad \text{on } \partial\Omega.$$

- When f and g are **proportional** and $f \geq 0$, $\partial\Omega$'s regularity follows from the one-phase Bernoulli problem's regularity theory (Alt and Caffarelli 1981; Weiss 1999; De Silva 2011; Caffarelli, Jerison, and Kenig 2004; Jerison and Savin 2015; De Silva and Jerison 2009).
- When f and g are **proportional** and f changes sign, it becomes a two-phase Bernoulli problem, with recent regularity results in Spolaor and Velichkov 2019; De Philippis, Spolaor, and Velichkov 2021.

The optimal shape Ω is **outwards-minimizing**, i.e.

$$-\int g(x)u_\Omega + |\Omega| \leq -\int g(x)u_A + |A|,$$

for every $A \supseteq \Omega$, which means that:

$$-\int f(x)u_\Omega + \frac{1}{c}|\Omega| \leq -\int f(x)u_A + \frac{1}{c}|A|.$$

Arguing as in Bucur, Mazzoleni, Pratelli, and Velichkov (2015), we find that u_Ω is **Lipschitz** and Ω is open.

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$$-\int f(x)u_\Omega + \frac{1}{C}|\Omega| \leq -\int f(x)u_\omega + \frac{1}{C}|\omega|.$$

By Bucur and Velichkov (2014), the function u_Ω is **non-degenerate**; in other words, there is $\eta > 0$ such that

$$\|u_\Omega\|_{L^\infty(B_r)} \leq \eta r \implies u \equiv 0 \text{ in } B_{r/2}.$$

1. Introduce the **adjoint function** v_Ω as a weak solution of $-\Delta v_\Omega = g$ on Ω, with $v_\Omega \in H_0^1(\Omega)$, so that:

$$\mathcal{F}(\Omega) = \int_D \left(\nabla u_\Omega \cdot \nabla v_\Omega - g(x)u_\Omega - f(x)v_\Omega + \chi_\Omega(x)\right)\, dx.$$

2. For any smooth vector field $\eta \in C_c^\infty(D;\mathbb{R}^d)$ and $\Omega_t = (\mathrm{Id} + t\eta)(\Omega)$, if Ω is optimal, then the first variation is zero: $\delta\mathcal{F}(\Omega)[\eta] = 0$.
3. A computation shows that u_Ω and v_Ω are solutions to the **free boundary system**:

$$\begin{cases} -\Delta u_\Omega = f \quad \text{and} \quad -\Delta v_\Omega = g & \text{in } \Omega, \\ u_\Omega = v_\Omega = 0 \quad \text{and} \quad |\nabla u_\Omega||\nabla v_\Omega| = 1 & \text{on } D \cap \partial\Omega. \end{cases} \tag{2}$$

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Definition. We say that (Ω, u, v) is a **viscosity solution** to

$$\begin{cases} \Omega = \{u > 0\} = \{v > 0\}, & \\ \Delta u = \Delta v = 0 & \text{in } \Omega \cap B_1, \\ G(|\nabla u|, |\nabla v|) = 1 & \text{on } \partial\Omega \cap B_1, \end{cases}$$

if $\forall x_0 \in \partial\Omega \cap B_1$ at which Ω admits a one-sided tangent ball there are

(a) a unit vector $\nu \in \mathbb{R}^d$;

(b) constants $\alpha, \beta > 0$ such that $G(\alpha, \beta) = 1$;

(c) a decreasing sequence of radii $r_k \to 0$;

such that the blow-up sequences u_{x_0, r_k} and v_{x_0, r_k} converge uniformly in every ball to the blow-up limits

$$u_0(x) := \alpha(x \cdot \nu)_+ \qquad \text{and} \qquad v_0(x) := \beta(x \cdot \nu)_+.$$

The functional $\int_{B_1} |\nabla U|^2 \, dx + |\{|U| > 0\} \cap B_1|$, where $U = (u, v)$, yields the following free boundary system:

$$\begin{cases} \Omega = \{u^2 + v^2 > 0\}, & \\ \Delta u = \Delta v = 0 & \text{in } \Omega \cap B_1, \\ |\nabla u|^2 + |\nabla v|^2 = 1 & \text{on } \partial\Omega \cap B_1. \end{cases}$$

- Flat free boundaries are $C^{1,\alpha}$ when $u, v > 0$ (Caffarelli, Shahgholian, and Yeressian 2018; Mazzoleni, Terracini, and Velichkov 2017; Kriventsov and F.H. Lin 2018);
- (Kriventsov and F.H. Lin 2019) any dimension, no sign assumption;
- (Mazzoleni, Terracini, and Velichkov 2020), flat NTA boundaries are $C^{1,\alpha}$; (De Silva and Tortone 2020), flat free boundaries are $C^{1,\alpha}$.

Theorem (Maiale, Tortone, and Velichkov (2023))

Let $u, v : B_1 \to \mathbb{R}$ be non-negative continuous functions, and $\Omega \subseteq B_1$ be any open set. Suppose that (Ω, u, v) is a solution to:

$$\begin{cases} \Omega = \{u > 0\} = \{v > 0\}, \\ -\Delta u = f \quad \text{and} \quad -\Delta v = g \quad \text{in } \Omega \cap B_1, \\ |\nabla u||\nabla v| = 1 \qquad\qquad\qquad \text{on } \partial\Omega \cap B_1. \end{cases}$$

In addition, assume that (u, v) is ϵ-flat for some $\epsilon \leq \epsilon_0$, that is,

$$(x \cdot \nu - \epsilon)_+ \leq u(x) \leq (x \cdot \nu + \epsilon)_+,$$
$$(x \cdot \nu - \epsilon)_+ \leq v(x) \leq (x \cdot \nu + \epsilon)_+,$$

in B_1. Then the free boundary $\partial\Omega$ is $C^{1,\alpha}$-regular in $B_{1/2}$.

Let $\Omega_t := (\mathrm{Id} + t\eta)(\Omega)$. If Ω is an optimal shape, then it is a **stable critical point** of the shape functional:

$$\frac{d}{dt}\Big|_{t=0} \mathcal{F}(\Omega_t) = \delta\mathcal{F}(\Omega)[\eta] = 0 \quad \text{and} \quad \frac{d^2}{dt^2}\Big|_{t=0} \mathcal{F}(\Omega_t) \geq 0.$$

This notion is stable under blow-up limits at $x_0 \in \partial\Omega \cap D$:

$$u_0(x) := \lim_{n\to+\infty} \frac{u_\Omega(x_0 + r_n x)}{r_n} \quad \text{and} \quad v_0(x) := \lim_{n\to+\infty} \frac{v_\Omega(x_0 + r_n x)}{r_n}.$$

So $\Omega_0 = \{u_0 > 0\}$ is also a stable critical point with $f \equiv g \equiv 0$. A Boundary Harnack Principle (Maiale, Tortone, and Velichkov 2022) implies

$$u_0/v_0 : \Omega_0 \to \mathbb{R}$$

is Hölder-continuous in Ω_0 up to the boundary.

A further blow-up in zero yields Ω_{00} stable critical point and $u_{00} = \Lambda v_{00}$; thus, u_{00} is a stable critical point for the Alt-Caffarelli's functional

$$\mathcal{G}(u, B_R) = \int_{B_R} |\nabla u|^2 \, dx + |B_R \cap \{u > 0\}| \qquad \text{in every } B_R \subset \mathbb{R}^d.$$

A monotonicity formula (Weiss 1999), in this context, is available when $p = 2$, and a third blow-up u_{000} is a **1-homogeneous stable critical point** for the Alt-Caffarelli's functional $\mathcal{G}$.

Thanks for your attention!

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- When $p > d$, existence follows under general assumptions as a consequence of *Poincaré inequality* and *Sobolev embedding*.
- When $p \leq d$, minimizing sequences Ω_n tend to split further and further, converging to a relaxed solution. This is generally not a domain, but a p-capacitary measure.
  When we further assume $f \in L^q(D)$, $q > d/p$, and a growth condition on the cost such as:

$$-g(x) \leq -C^{-1} f(x) \qquad \text{for every } x \in D,$$

  then the measure corresponds to an open set, solution of the initial problem. **The summability assumption on f cannot be removed**.
- Finite perimeter is obtained via a strategy from (Bucur 2012) developed for the eigenvalue problem.

The proof of $\mathcal{H}^{d-d^*+\varepsilon}(\mathrm{Sing}(\partial\Omega)) = 0$ proceeds as follows:

1. The bounds on d^* for minimizers of $\mathcal{G}$ rely on the well-known CJK stability inequality Caffarelli, Jerison, and Kenig 2004. However, this inequality is challenging to apply when passing to blow-up limits and developing a dimension reduction principle.
2. Our notion of **stable critical point** allows to develop a dimension reduction principle and that there is a critical dimension d^*.
3. Our stability notion is equivalent to the stability inequality on smooth cones (cones with isolated singularities), leading to the bounds $d^* \geq 5$ De Silva and Jerison 2009 and $d^* \leq 7$ Jerison and Savin 2015.

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